GURDHARAM JEET SINGH

Full-Stack Software Engineer • Web Systems & Cloud Automation

Bathinda / Muktsar, Punjab, India • +91 90411 72159 • gurination1@gmail.com • github.com/gurination1 • linkedin.com/in/gurdharam • gurdharam.com

TECHNICAL SKILLS

Languages & Core: TypeScript, JavaScript (ES6+), Python, Go, SQL (PostgreSQL, MariaDB), HTML5, CSS3, Linux/Bash

Frontend & Spatial Web: React 19, Next.js 15 (App Router, Server Components), Tailwind CSS 4, Vite 6, Three.js, WebGL2, Canvas API, Framer Motion, Lenis Scroll, Core Web Vitals (LCP, CLS < 0.02, FCP)

Backend & Distributed Systems: Node.js, Express, FastAPI, PostgreSQL, Supabase, MariaDB, Redis, BullMQ, WebSockets, REST APIs, Docker

Cloud, AI & Infrastructure: GitHub Actions 24/7 CI/CD Fleets, Meta Cloud API (WABA), Gemini AI Multimodal, FFmpeg, Railway, AWS (S3, Lambda), Exotel Telephony, Silero VAD, Sarvam AI, Groq LPU

WORK EXPERIENCE

CP Systems **2024 – 2025**

Software Developer • Full-Stack Services & Internal Tooling *Remote / Punjab, India*

- **High-Throughput Services & SQL Profiling:** Engineered internal REST services in Node.js and TypeScript, optimizing relational database query plans and indexing strategies to reduce p95 query latency by 42% across critical workflow endpoints.
- **CI/CD Test Automation & Production Reliability:** Implemented GitHub Actions CI/CD test automation pipelines with composite caching, cutting build validation time from 14m to 3.8m while enforcing automated test gates across sprint deployments.

Gurdharam Enterprise **2025 – Present**

Full-Stack Software Engineer & Solutions Engineer *Remote / Punjab, India*

- **Commercial Client Delivery & WebGL Platforms:** Architected production web platforms, agency portals (Branders), and interactive 3D WebGL interfaces for clients, achieving consistent sub-800ms LCP and 0.00 CLS across multi-region edge CDNs.
- **Real-Time Conversational Telephony AI:** Built low-latency telephony pipeline over Exotel full-duplex WebSockets, Silero VAD barge-in interruption, and streaming Sarvam AI Indic LLMs (<700ms turn latency), alongside resilient Meta Cloud API gateways.

KEY PRODUCTION PLATFORMS & SYSTEMS

SOLUM Minerals | *Next.js 15, Three.js, WebGL2, OffscreenCanvas, Tailwind CSS* **2026 (Oct) – Present** | Open Source

- **Procedural In-Memory Asset Synthesis:** Engineered procedural Three.js asset pipeline synthesizing dynamic 1024x1024 noise textures in-memory via Canvas API, eliminating 15MB of static 3D mesh downloads and cutting initial bundle payload by 85%.
- **Decoupled Kinematics & Web Vitals:** Decoupled Three.js render loops from DOM scroll events via normalized lerp interpolation in Next.js 15 App Router, maintaining 60 FPS viewport rendering while achieving CLS < 0.01 and 99+ Lighthouse performance.

yt-auto | *Python, FFmpeg, GitHub Actions CI/CD Matrix, Gemini Multimodal AI* **2025 – Present** | Open Source

- **Event-Driven Cloud Processing Fleet:** Architected media processing fleet running multi-channel batch rendering across parallelized GitHub Actions matrix workers, orchestrating 150+ automated video builds weekly with zero dedicated server overhead.
- **Multimodal QA Gate & Filtergraph Mastering:** Automated video synthesis and audio mastering via custom FFmpeg filtergraphs with sidechain compression, gated by a Gemini Multimodal AI evaluation pipeline rejecting visual artifacts (≥85 score threshold).

EduCore | *React 19, TypeScript, Express, PostgreSQL, MariaDB, Railway* **2025 – Present** | Open Source

- **Dual-Engine Database Abstraction Layer:** Engineered multi-tenant campus ERP supporting pooled PostgreSQL and MariaDB instances with connection-drain health checks, connection recovery, and sub-15ms p90 query latency under concurrent load.
- **Declarative RBAC & Audit Logging:** Built declarative role-based access control engine across 9 authorization tiers, coupled with immutable change-data-capture audit logging recording state diffs, actor IP, and device profiles.

Dream Heights | *React 18, TypeScript, Tailwind CSS, Framer Motion, Lenis Smooth Scroll* **2025 – Present** | Open Source

- **Spatial Navigation & Coordinate Mapping:** Architected luxury residence exploration portal in React 18 and Vite 6 with dynamic SVG coordinate mapping, modular floor-plan modals, and lazy-loaded architectural asset bundles.
- **GPU Composite Layers & Frame Budgets:** Optimized client-side frame budgets using GPU composite layers (transform3d) and virtualized scroll interpolation, maintaining stable 60 FPS refresh rates across high-DPI viewports without layout thrashing.

EDUCATION & CREDENTIALS

Baba Farid Group of Institutions (BFGI) **Expected May 2027**

Bachelor of Science (B.Sc.) in Computer Science • Coursework: Data Structures & Algorithms, DBMS, Networks, OS *Punjab, India*

Credentials & Verification: Meta Verified Business Entity • Official WhatsApp Business Platform Developer • Sarvam AI Startup Partner • Govt. MSME Registered Enterprise (UDYAM-PB-03-0079744) • Additional Production Systems: *SnapAR Studio (Camera Kit WebGL2)*, *TakeMyInterview AI*, *Fasal Doctor (Patent Ref: IDF-FD-2026-0413)*.